

An indication name containing a resistance or refractory-state term predicts higher repurposing success.

AUDIT NUMBERS - FROM THE REGISTRY AT SAVE TIME

FRAME	TEST	OR	95% CI	N	DISCOVERY RAW P	FDR Q	CONFIRM RAW P
narrow	binary	-	-	-	-	-	-
broad	fisher_exact	0.0931	[0.0279, 0.441]	4203	3.57e-4	1.25e-3	4.63e-6

CONFOUND	UNADJ. OR	ADJ. OR	ADJ. 95% CI	ADJ. P	SURVIVES?
oncology therapeutic-area concentration	0.0931	0.0609	[0.0124, 0.2999]	5.80e-4	survives
prior repurposing momentum	0.0931	0.104	[0.0148, 0.7334]	2.31e-2	survives
established-product incumbency	0.0931	0.0892	[0.0193, 0.4116]	1.95e-3	survives

NARRATIVE - CLAUDE OPUS 4.8

Drug-Repurposing Statistical Finding: Audit Report

Hypothesis ID: run-9e893c07-H21

Hypothesis

The hypothesis, stated verbatim, is: "An indication name containing a resistance or refractory-state term predicts higher repurposing success." It originates from the source domain of **collateral sensitivity in resistance rotation** and was proposed by the model **Sol**. No mechanistic justification was supplied in FACTS (the field is empty).

Statistical evidence underneath the p-value

Two outcome framings are present in FACTS: **narrow** and **broad**. The narrow framing carries no computable statistics — its discovery and confirmation p-values are NaN, its FDR q-value is null, and its effect_size object is null — so no effect summary can be reported for it. Only the broad framing yields a computable effect.

Framing	Test type	Odds ratio	95% CI	n	Discovery raw p	FDR q
narrow	binary	null (NaN)	null	null	NaN	null
broad	fisher_exact	0.0931	0.0279 – 0.4414	2030.00035735340306005420.00125073691071019		

narrow: No effect size, sample size, or usable p-value is available in FACTS; nothing can be stated about the direction or magnitude of this framing.

broad: The odds ratio of 0.0931 is well below 1, meaning the presence of a resistance/refractory term in the indication name is associated with *lower*, not higher, odds of repurposing success in this dataset — an odds reduction of roughly an order of magnitude (odds about one-tenth), which runs opposite to the hypothesis's stated prediction of "higher" success.

Discovery vs. confirmation

Framing	Discovery raw p	FDR q	Discovery pass	Confirmation raw p	Confirmation pass
narrow	NaN	null	false	NaN	false
broad	0.00035735340306005420.00125073691071019	true	4.630066606303871e-06	true	

Only the broad framing passed both stages. Passing the confirmation test means the effect **reproduced on a holdout split of data that was never used during the discovery phase**, so the association is not an artifact of the particular data used to originally detect it. The narrow framing failed both discovery and confirmation and is not a supported result.

Confound checks

The confound_check object has status `completed` and carries the note: "*candidate explanations checked — not the confirmed mechanism.*" Three alternative explanations were tested, all of them computable from the dataset. The unadjusted primary effect against which each was compared is OR = 0.0931, p = 0.000357, n = 4203.

1. Oncology therapeutic-area concentration

Rationale: Resistance/refractory/relapsed terminology is overwhelmingly concentrated in oncology and hematology indications, which have unusually dense repurposing pipelines and high line-of-therapy churn independent of any evolutionary trade-off; the keyword may thus act as a proxy for "is an oncology indication," with oncology's baseline activity driving the signal.

Computable: Yes.

Unadjusted: OR = 0.0931, p = 0.000357.

Adjusted: OR = 0.0609, 95% CI 0.0124 – 0.2999, p = 0.00058.

Survives adjustment: **Yes** (note: "effect survives adjustment").

2. Prior repurposing momentum

Rationale: Refractory/relapsed indications are by definition later-line settings that only arise after prior use, so they correlate mechanically with high prior_repurposing_count; drugs with many prior attempts are structurally more likely to log another success, potentially creating the correlation without the hypothesis being true.

Computable: Yes.

Unadjusted: OR = 0.0931, p = 0.000357.

Adjusted: OR = 0.1042, 95% CI 0.0148 – 0.7334, p = 0.02313.

Survives adjustment: **Yes** (note: "effect survives adjustment").

3. Established-product incumbency

Rationale: Resistance/refractory indications typically presuppose an existing first-line established product that the population has failed, so the keyword co-occurs with established-product drugs, which have marketing infrastructure, safety data, and label-expansion pathways that inflate repurposing success independently of any new biological vulnerability.

Computable: Yes.

Unadjusted: OR = 0.0931, p = 0.000357.

Adjusted: OR = 0.0892, 95% CI 0.0193 – 0.4116, p = 0.001951.

Survives adjustment: **Yes** (note: "effect survives adjustment").

Verdict: All three tested confounds were computable, and the primary effect (broad framing) survived adjustment for every one of them: oncology concentration (adjusted OR 0.0609, p 0.00058), prior repurposing momentum (adjusted OR 0.1042, p 0.02313), and established-product incumbency (adjusted OR 0.0892, p 0.001951). No computable confound caused the effect to fail. The weakest post-adjustment result was for prior repurposing momentum, where the adjusted p-value (0.02313) is the largest of the three and the upper CI bound (0.7334) approaches 1, though it remains below 1 and below the 0.05 threshold.

Bottom line

In this curated dataset, the presence of a resistance/refractory term in an indication name is robustly associated with an odds ratio of 0.0931 for repurposing success (broad framing, n = 4203) — an association that passed FDR correction, reproduced on an independent holdout, and survived adjustment for all three tested confounds. Note that the direction of this association is *lower* odds of success, opposite to the hypothesis's prediction of "higher" success, and this is a statistical association only — it is not evidence of causation, biological mechanism, or clinical relevance (the confound module itself flags this as "not the confirmed mechanism"). The strongest remaining caveat is that the untested space of confounders is unbounded and that even among tested ones, prior repurposing momentum weakened the effect substantially (adjusted p = 0.02313, upper CI 0.7334), so the margin against this particular alternative explanation is comparatively thin.